GSE 518
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Suggested Solutions to Problem Set 7

1. We first prove that $E\left(\overline{X}-\overline{Y}\right)=\mu_x-\mu_y$:

$$\begin{aligned}
E\left(\overline{X}-\overline{Y}\right) &= E\left(\overline{X}\right)-E\left(\overline{Y}\right)=E\left(\frac{1}{n}\sum_{i=1}^{n}X_i\right)-E\left(\frac{1}{m}\sum_{i=1}^{m}Y_i\right) \\
&= \frac{1}{n}E\left(\sum_{i=1}^{n}X_i\right)-\frac{1}{m}E\left(\sum_{i=1}^{m}Y_i\right)=\frac{1}{n}\sum_{i=1}^{n}E\left(X_i\right)-\frac{1}{m}\sum_{i=1}^{m}E\left(Y_i\right) \\
&= \frac{1}{n}\sum_{i=1}^{n}\mu_x-\frac{1}{m}\sum_{i=1}^{m}\mu_y=\frac{n\mu_x}{n}-\frac{m\mu_y}{m}=\mu_x-\mu_y
\end{aligned}$$

as required.
We now show $Var\left(\overline{X}-\overline{Y}\right)=\frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}$:

$$\begin{aligned}
Var\left(\overline{X}-\overline{Y}\right) &= Var\left(\overline{X}\right)+Var\left(\overline{Y}\right)-2cov\left(\overline{X},\overline{Y}\right) \\
&= Var\left(\frac{1}{n}\sum_{i=1}^{n}X_i\right)+Var\left(\frac{1}{m}\sum_{i=1}^{m}Y_i\right)-2Cov\left(\frac{1}{n}\sum_{i=1}^{n}X_i,\frac{1}{m}\sum_{j=1}^{m}Y_j\right) \\
&= \frac{1}{n^2}Var\left(\sum_{i=1}^{n}X_i\right)+\frac{1}{m^2}Var\left(\sum_{i=1}^{m}Y_i\right)-2\frac{1}{n}\frac{1}{m}Cov\left(\sum_{i=1}^{n}X_i,\sum_{j=1}^{m}Y_j\right) \\
&= \frac{1}{n^2}\sum_{i=1}^{n}Var\left(X_i\right)+\frac{1}{m^2}\sum_{i=1}^{m}Var\left(Y_i\right)-2\frac{1}{n}\frac{1}{m}\sum_{i=1}^{n}\sum_{j=1}^{m}Cov\left(X_i,Y_j\right) \\
&= \frac{1}{n^2}\sum_{i=1}^{n}\sigma_x^2+\frac{1}{m^2}\sum_{i=1}^{m}\sigma_y^2=\frac{n\sigma_x^2}{n^2}+\frac{m\sigma_y^2}{m^2}=\frac{\sigma_x^2}{n}+\frac{\sigma_y^2}{m}
\end{aligned}$$

In the third line we use the fact that $Cov\left(aX,bY\right)=abCov\left(X,Y\right)$ for any constants a and b. In the fourth line we use the fact that $X_1,X_2,\ldots,X_n$ is a random sample, so by independence we have that $Var\left(\sum_{i=1}^{n}X_i\right)=\sum_{i=1}^{n}Var\left(X_i\right)$ (i.e., all covariance terms are zero). The same applies for the term $Var\left(\sum_{i=1}^{m}Y_i\right)$. In this line we also used the fact shown in a previous problem set (problem set 6, question 3) that $Cov\left(X+W,Y+Z\right)=Cov\left(X,Y\right)+Cov\left(X,Z\right)+Cov\left(W,Y\right)+Cov\left(W,Z\right)$. Finally, in the last line we used the fact that $X_1,X_2,\ldots,X_n$ and $Y_1,Y_2,\ldots,Y_m$ are two independent random samples, so that $Cov\left(X_i,Y_j\right)=0$ for all i and j.

2. In this question we use the same properties of the covariance we used in question 1. In this case we have

$$\begin{aligned} Cov\left(X_i - \overline{X}, \overline{X}\right) &= Cov\left(X_i, \overline{X}\right) - Cov\left(\overline{X}, \overline{X}\right) \\ &= Cov\left(X_i, \frac{1}{n}\sum_{i=1}^{n} X_i\right) - Var\left(\overline{X}\right) \\ &= \frac{1}{n} Cov\left(X_i, \sum_{i=1}^{n} X_i\right) - Var\left(\overline{X}\right) \\ &= \frac{1}{n} Cov\left(X_i, X_i\right) + \frac{1}{n}\sum_{k \neq i} Cov\left(X_i, X_k\right) - Var\left(\overline{X}\right) \\ &= \frac{1}{n} Cov\left(X_i, X_i\right) - Var\left(\overline{X}\right) \\ &= \frac{1}{n} Var\left(X_i\right) - Var\left(\overline{X}\right) \\ &= \frac{1}{n}\sigma^2 - \frac{\sigma^2}{n} = 0 \end{aligned}$$

In the second line we use the fact that $Cov\left(U, U\right) = Var\left(U\right)$. In the fifth line we used the fact that since $X_1, X_2, \ldots, X_n$ is a random sample, then the Xs are mutually independent, so $Cov\left(X_i, X_k\right) = 0$ for $i \neq k$ and therefore $\frac{1}{n}\sum\limits_{k \neq i} Cov\left(X_i, X_k\right) = 0$.

3. Since the random sample $X_1, X_2, \ldots, X_n$ comes from a normal distribution $N\left(\mu, \sigma^2\right)$, Theorem 5.3.1 holds and $\left(n-1\right)s^2/\sigma^2$ follows a Chi-square distribution with $n-1$ degrees of freedom. Note that if the random sample does not come from a normal distribution Theorem 5.3.1 does not hold in general.
Remember that by definition $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2$. From Theorem 5.3.1 part (c) we know that $\left(n-1\right)s^2/\sigma^2$ is distributed as a Chi-square with $n-1$ degrees of freedom, or $\chi^2_{(n-1)}$. Similarly, we know that the variance of a $\chi^2_{(r)}$ equals $2r$. Therefore, we have that $Var\left(\left(n-1\right)s^2/\sigma^2\right) = 2\left(n-1\right)$. Hence we have

$$\begin{aligned} 2\left(n-1\right) &= Var\left(\frac{\left(n-1\right)s^2}{\sigma^2}\right) \\ &= \frac{\left(n-1\right)^2}{\sigma^4} Var\left(s^2\right) \end{aligned}$$

which implies

$$Var\left(s^2\right) = \frac{2\left(n-1\right)\sigma^4}{\left(n-1\right)^2} = \frac{2\sigma^4}{n-1}$$

as required.

4. Using theorem 5.3.1 in CB part (c) (note that the random sample comes from a normal

distribution), we have that $\frac{(n-1)s_x^2}{\sigma_x^2}$ follows a Chi-square distribution with $n-1$ degrees of freedom, or $\chi^2_{(n-1)}$. Similarly, we have that $\frac{(m-1)s_y^2}{\sigma_y^2}$ follows a Chi-square distribution with $m-1$ degrees of freedom, or $\chi^2_{(m-1)}$. Moreover, since $X_1, X_2, \ldots, X_n$ and $Y_1, Y_2, \ldots, Y_m$ are two independent random samples, we have that s_x^2 and s_y^2 are also independent. Hence we have that

$$\frac{s_x^2/\sigma_x^2}{s_y^2/\sigma_y^2} = \frac{\frac{(n-1)s_x^2}{\sigma_x^2}\frac{1}{(n-1)}}{\frac{(m-1)s_y^2}{\sigma_y^2}\frac{1}{(m-1)}} \sim \frac{\chi^2_{(n-1)}/(n-1)}{\chi^2_{(m-1)}/(m-1)} = F_{n-1,m-1}$$

That is, $\frac{s_x^2/\sigma_x^2}{s_y^2/\sigma_y^2}$ follows the same distribution as the ratio of two independent Chi-square random variables divided by their corresponding degrees of freedom, which by "Result 2" from class (or Definition 5.3.6 in CB) follows an F distribution with $n-1$ and $m-1$ degrees of freedom. (For more on this problem see Example 5.3.5 and Definition 5.3.6 in CB).

5. In this problem we have that $X_i, i = 1, 2, 3$ are independent from $N(i, i^2)$ distributions.
(a) Since $X_i \sim N(i, i^2)$, we have that $\left(\frac{X_i - i}{i}\right) \sim N(0, 1)$ distribution. Remember that the square of a random variable which follows a standard normal distribution follows a Chi-square with one degree of freedom, or $\chi^2_{(1)}$. Moreover, remember that the sum of independent Chi-square random variables follows a Chi-square distribution with degrees of freedom equal to the sum of the degrees of freedom (e.g., see Lemma 5.3.2 in CB). Hence, given that the X_i are independent $N(i, i^2)$, we have that

$$\sum_{i=1}^{3} \left(\frac{X_i - i}{i}\right)^2 \sim \chi^2_{(3)}$$

(b) Remember that a t-distribution with r degrees of freedom is obtained as the ratio of a standard-normally distributed random variable and the square root of a Chi-square random variable with r degrees of freedom divided by r, where both random variables are independent of each other (see "Result 1" from class or Definition 5.3.4 in CB). Informally

$$t_{(r)} = \frac{N(0, 1)}{\sqrt{\chi^2_{(r)}/r}}$$

Hence in this problem we need to find a random variable that follows a $N(0, 1)$ and another which follows a $\chi^2_{(2)}$ and is *independent* of the first one. From part (a) we know, for example, that $\left(\frac{X_1 - 1}{1}\right) = X_1 - 1 \sim N(0, 1)$. Similarly, we have that $\left(\frac{X_2 - 2}{2}\right) \sim N(0, 1)$ and $\left(\frac{X_3 - 3}{3}\right) \sim N(0, 1)$, so using the same arguments as in part (a) we have that $\sum_{i=2}^{3} \left(\frac{X_i - i}{i}\right)^2 \sim \chi^2_{(2)}$. Therefore, we have that

$$\frac{X_1 - 1}{\sqrt{\frac{\sum_{i=2}^{3}\left(\frac{X_i - i}{i}\right)^2}{2}}} \sim t_{(2)}$$

It is important to note that we cannot use the same variable (say, X_1) in the numerator and denominator because in that case the variables would not be independent.

(c) By the properties of the t and F distributions, we know that the square of a random variable which follows a $t_{(r)}$ distribution follows an $F_{1,2}$ distribution (e.g., Theorem 5.3.8 in CB). Hence,

$$\frac{(X_1 - 1)^2}{\frac{\sum_{i=2}^{3}\left(\frac{X_i - i}{i}\right)^2}{2}} \sim F_{1,2}$$